Friday cardio day — court & conditioning

Knee-safe · hypermobility-adapted · ~55–63 min · badminton-focused

⚠ Hypermobility rules
- Never lock out joints — soft bend always
- No lower body static stretching
- Stability & control over flexibility
- Avoid deep lunges, squats, end-range
- Isometric holds > static stretches

🦵 Knee protection rules
- Minimize high-impact landings
- No explosive jump-and-land movements
- Wear knee sleeves if available
- Pain = STOP immediately
- Brake with muscles, not joints

Phase 1 — Activation & warm-up (~10–12 min)

Band activation circuit — 2 rounds

1	**Banded clamshells** Glute medius — knee stabilizer #1	12 reps/side Band
2	**Banded lateral walks (don't let knees cave)** Hip abductors — lateral knee stability	10 steps/dir Band
3	**Banded terminal knee extensions** (anchor behind knee · extend from slight bend to straight) VMO — kneecap tracking muscle	12 reps/leg Band
4	**Single-leg mini balance hold (slight knee bend)** Proprioception — essential for hypermobile joints	15s / leg Bodyweight

Movement prep — 4–5 min

Walking knee hugs (gentle) → Lateral shuffles (controlled pace) →
High knees (reduced height) → Split-step practice (soft landings, absorb with bent knees)

Phase 2 — Badminton court work (~25–28 min)

Block A — Modified footwork (7–8 min)

5	**4-corner shadow drill (skip deep net lunges)** Move at 70–80% speed · smooth deceleration into each corner Front-right, front-left, back-right, back-left *Cue: "Brake with your muscles, not your joints"*	3 × 1 min 45s rest
6	**Multi-shuttle feeding drill (if partner available)** Continuous retrieval at pace from center position	2 × 2 min

Block B — Rally-based conditioning (12–15 min)

7	**Continuous rallies — clears & drops over full smashes** Limit smashes to 2–3 per rally (heavy landing knee load) Focus: court positioning & shot placement over raw power *If knees ache → switch to stationary feeding drills from center*	3–4 rounds 3 min on / 1 min rest

Block C — Modified match play (5–6 min)

8	**Short set at ~80% controlled intensity** No diving · no full split-step lunges to net · no jump smashes Competitive enough for match stamina — controlled for knee safety	Race to 11 Full rules

Phase 3 — Complementary conditioning (~10–12 min · rotate weekly · pick ONE)

A	**Cycling — stationary or outdoor (DEFAULT)** Zero knee impact · builds quad endurance around joint Fully controllable resistance · low-impact aerobic base	10–12 min Moderate pace ★ MOST WEEKS
B	**Swimming — freestyle or backstroke only** Zero impact · decompresses joints · full-body aerobic ⚠ Avoid breaststroke kick — stresses inner knee	10–12 min Easy–moderate
C	**Jump rope — GOOD-KNEE-DAY ONLY** Stay on balls of feet · low jump height · calf/ankle conditioning ⚠ Involves impact — skip if any knee discomfort that day	4 × 45s on 45s rest

Phase 4 — Stabilization cool-down (~8–10 min · replaces traditional stretching)

Hypermobile joints need control, not more range — isometric holds replace static stretches

Heart rate recovery — 2 min

Walk or slow cycling to bring heart rate down gradually

Isometric & stability circuit — 6–8 min

9	**Wall sit hold (knees at ~90° · don't slide deeper)** Quad endurance around knee joint without impact	30s × 2
10	**Single-leg stance — eyes closed** Proprioception training — critical for hypermobile joints	20s / leg
11	**Banded glute bridge hold** Hip stability · takes stress off knees	20s × 2 Band
12	**Plank** Core stability — supports all movement patterns	30s

Active upper body cool-down — replaces static stretches (~2 min)

13	**Slow eccentric push-up — 8s lowering (from knees OK)** Pec major · anterior delt · triceps — lengthens chest under load Replaces passive chest doorway stretch · from upper body mobility protocol S2	2 reps 8s eccentric each
14	**Shoulder CARs — slow recovery tempo** Full shoulder capsule · rotator cuff — active scan replaces passive cross-body stretch From upper body mobility protocol S1 · compare feel to pre-session	2 each direction × each arm ~45s per circle
15	**Active neck rotation with isometric resistance** Upper traps · SCM · cervical stabilizers — replaces passive neck side-bends Hand on side of head · push head into hand ~20% effort · hold 5s	2 reps/side 5s hold each

❌ SKIP: All passive stretches — excessive range increases instability risk for hEDS joints

🍽 Post-session fueling
Shake at venue → light meal at home

💡 Friday lunch tip
Slightly more carbs OK — you'll burn them

Progression path — weeks 1–6+

Weeks 1–2 · Foundation
- 70% court intensity
- Default to cycling
- Learn band activation circuit form
- Build movement habits

Weeks 3–4 · Building
- 80% court intensity
- Try jump rope on good-knee days
- +5s to wall sits
- More rally duration

Weeks 5–6+ · Full
- Reintroduce 6-corner if pain-free
- Allow more smashes
- Full intensity if 2+ weeks pain-free
- Keep band warm-up

🚨 Red flags — stop & reassess

- Sharp pain (not dull discomfort) in either knee during or after
- Knee swelling the day after
- Feeling of "giving way" or instability mid-movement
- Pain that gets worse as the session progresses (not better)

→ Rest that week's cardio · consider physio check-in before resuming

💡 Equipment recommendation: knee compression sleeves
Proprioceptive feedback helps hypermobile joints know where they are in space

Every session — remember
- Band warm-up is permanent maintenance — not just rehab
- Saturday foam rolling complements this session
- Same post-workout pattern: shake at venue, light meal at home
- One cardio day won't interfere with Mon/Wed/Fri lifting
- No need to add a second day unless recovery supports it